

# **Complete Azure Hub & Spoke Landing Zone + Terraform + DevSecOps (35 Points Guide)**

## **1. Requirement Gathering & Planning**

- Define Business Requirements
- Define Security & Compliance Requirements
- Define RTO/RPO
- Identify Monolithic & Microservices Applications

## **2. Azure Landing Zone Design**

- Hub Network for Shared Services
- Spoke Networks for Applications
- Environment Isolation Strategy

## **3. Subscription Strategy**

- Management Subscription
- Connectivity Subscription
- Identity Subscription
- Production & Non-Production Subscription

## **4. Management Group Hierarchy**

- Tenant Root Group
- Platform Management Groups
- Landing Zone Hierarchy
- Sandbox Management Group

## **5. Governance & Compliance**

- Azure Policy
- Initiatives
- RBAC
- Budget Alerts
- Resource Locks

## **6. IAM Architecture**

- Microsoft Entra ID
- PIM

- Conditional Access
- Managed Identity
- Least Privilege RBAC

## **7. Networking Architecture**

- Azure Firewall Premium
- VPN Gateway
- Azure Bastion
- Private DNS
- NSGs & UDRs

## **8. Terraform Repository Structure**

- modules/
- environments/
- pipelines/
- scripts/

## **9. Parent & Child Modules**

- Parent Modules for orchestration
- Child Modules for reusable components
- Environment-specific deployments

## **10. Generic Terraform Module Design**

- `map(object())`
- dynamic blocks
- `for_each`
- conditional expressions
- locals
- optional attributes

## **11. `map(object())` Example**

- Reusable VNet & Subnet Objects
- Environment Driven Configurations

## **12. Dynamic Block Example**

- Dynamic Subnet Creation
- Dynamic NSG Rules

- Reusable Generic Logic

## **13. Conditional Logic**

- Conditional Resources
- Enable/Disable Public IPs
- Environment-Based Deployment

## **14. Optional Attributes**

- Optional Availability Zones
- Flexible Configurations
- Simplified Module Design

## **15. Remote State Management**

- Azure Storage Backend
- Blob Container
- Remote State Locking
- Environment Isolation

## **16. CI/CD Strategy**

- Terraform Validate
- Terraform Plan
- Approval Gates
- Terraform Apply

## **17. Trunk-Based Branching Strategy**

- main branch
- Short-lived feature branches
- Pull Request workflow

## **18. GitHub vs Azure Repos**

- GitHub Actions Integration
- Azure Repos Governance
- OIDC Federation

## **19. GitHub Actions Pipeline**

- Lint
- Validate

- Security Scan
- Apply

## **20. Azure DevOps Pipeline**

- Validate
- Plan
- Approval
- Apply
- Drift Detection

## **21. DevSecOps Security Integration**

- Checkov
- Tfsec
- SonarQube
- Trivy
- SAST & DAST

## **22. Secret Management**

- Azure Key Vault
- Managed Identity
- OIDC Authentication
- No Hardcoded Secrets

## **23. AKS Architecture**

- Private AKS
- Azure CNI
- Autoscaling
- Node Pools
- Network Policies

## **24. Monolithic Application Deployment**

- App Gateway
- App Service / VMSS
- Database Layer
- Autoscaling

## **25. Microservices Architecture**

- AKS
- Docker
- Helm
- ArgoCD
- Istio

## **26. Monitoring & Observability**

- Azure Monitor
- Application Insights
- Prometheus
- Grafana
- Centralized Logging

## **27. Backup Strategy**

- Azure Backup
- Recovery Services Vault
- Immutable Backup
- Cross-region Backup

## **28. Disaster Recovery Strategy**

- Active-Passive
- Active-Active
- Traffic Manager
- Front Door

## **29. Security Best Practices**

- Defender for Cloud
- Sentinel
- Private Link
- TLS 1.2+
- Disk Encryption

## **30. Cost Optimization**

- Reserved Instances
- Autoscaling
- Spot VMs
- Rightsizing

## **31. Logging & Audit**

- Diagnostic Settings
- Activity Logs
- Retention Policies
- SIEM Integration

## **32. Production Readiness Checklist**

- HA Configured
- DR Configured
- Monitoring Enabled
- Security Scanning Enabled

## **33. End-to-End Flow**

- Developer → Repo → PR → Validate → Scan → Plan → Approval → Apply

## **34. Enterprise-Level Best Practices**

- Reusable Generic Modules
- Policy as Code
- GitOps
- Immutable Infrastructure
- Zero Trust Security

## **35. Advanced DevSecOps Add-ons & Tech Stack**

- MCP AI Agents
- AIOps
- SRE Automation
- Chaos Engineering
- Blue-Green & Canary Deployment

# Recommended Enterprise Tech Stack

| Category       | Recommended Tools                            |
|----------------|----------------------------------------------|
| Infrastructure | Terraform, Azure Landing Zones, Bicep        |
| CI/CD          | GitHub Actions, Azure DevOps                 |
| Containers     | Docker, AKS, Helm, ArgoCD                    |
| Monitoring     | Prometheus, Grafana, Azure Monitor           |
| Security       | Defender for Cloud, Sentinel, Checkov, Tfsec |
| Service Mesh   | Istio                                        |
| GitOps         | ArgoCD, FluxCD                               |
| Observability  | Log Analytics, Application Insights          |
| Backup & DR    | Recovery Services Vault, Front Door          |